

Armaan Sandhu

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Education

Worcester Polytechnic Institute <i>Worcester, MA</i>	2025 - Present
Computer Science GPA: 4.0 Dean's List (Fall 2025, Spring 2026) Expected May 2029	
Carmel Junior College <i>Jamshedpur, India</i>	2010 - 2025
High School Diploma 97% (Unweighted) Valedictorian, Class of 2025	

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript

Tools: Git, GDB, Linux, Bash, Valgrind

Concepts: Systems Programming, Concurrent/Async Programming, Binary Analysis, Query Planning, AI Infrastructure, Open Source Development

Open Source Contributions

lance/lancedb (rust)	May 2026 - June 2026
- Implemented native DataFusion expression support across Lance and LanceDB, removing SQL serialization overhead in programmatic query planning and enabling efficient predicate construction. - Improved async embedding execution by isolating blocking workloads in a dedicated executor, preventing thread-pool contention under concurrent search workloads. - Added table and query engine improvements including rename-table support and merge-insert execution enhancements.	
Lightpanda Browser Engine (Zig)	May 2026 - June 2026
Implemented the W3C File API in a browser engine, adding standards-compliant file objects, metadata handling, MIME-type validation, and integration tests.	
CPython	May 2026
Fixed build-system issues in the Cygwin port of CPython, improving compiler portability, static library handling, and platform configuration correctness.	
LiteLLM Proxy (Python)	June 2026
- Fixed distributed budget-accounting and enforcement issues in proxy systems, including resolution of phantom budget reservations across Redis-backed deployments. - Corrected authorization behavior for model-discovery endpoints, ensuring compatibility with OpenAI-style clients without weakening inference budget enforcement.	

Experience

Open Source Author & MCP Deployment Consultant <i>mcp-persist / MCP Python SDK deployments</i>	May 2026 - Present
- Built and maintain mcp-persist, a production-grade persistence layer for MCP deployments supporting Redis, SQLite, and PostgreSQL backends: actively deployed in production MCP server environments. - Designed an async event-store architecture with 200+ tests and CI coverage across Python 3.10-3.13. - Support real-world MCP deployments by guiding integration, migration, and backend selection across production environments. - Contribute upstream fixes to MCP Python SDK spanning async runtime correctness, CI stability, and type safety improvements.	
Peer Learning Assistant <i>Computer Science Department, WPI</i>	August 2026 - Present
Supported undergraduate students in CS coursework through one-on-one debugging sessions, code review, and targeted explanations of systems-level concepts. Guided peers toward independent problem-solving rather than direct solutions.	
Community Advisor <i>Office of Academic Advising, WPI</i>	August 2026 - Present
Managed a residential student community and served as a frontline resource for academic and personal challenges. Helped students navigate course planning, campus resources, and the transition to college life.	

Awards & Recognition

- Valedictorian** | Carmel Junior College (Class of 2025)
- Most Outstanding Student** | Awarded for excellence in academics, leadership, and extracurriculars.
- Interschool Java Coding Competition** | 2nd Place (2023, 2024)
- Ni Dan Black Belt** | Shotokan Karate India (IOC-recognized federation)